gene, clarifying the clinical diagnosis and altering treatment for the patient (see Chapter 16).
- Scientists in the pharmaceutical industry sequence cancer cell DNA to identify specific changes in oncogenic signaling pathways inappropriately activated by a somatic mutation, leading to the development of specific inhibitors that reliably induce remissions of the cancers in patients (Case 10).

Categories of Genetic Disease

Virtually any disease is the result of the combined action of genes and environment, but the relative role of the genetic component may be large or small. Among disorders caused wholly or partly by genetic factors, three main types are recognized: chromosome disorders, single-gene disorders, and multifactorial disorders.

In **chromosome disorders,** the defect is due not to a single mistake in the genetic blueprint but to an excess or a deficiency of the genes located on entire chromosomes or chromosome segments. For example, the presence of an extra copy of one chromosome, chromosome 21, underlies a specific disorder, Down syndrome, even though no individual gene on that chromosome is abnormal. Duplication or deletion of smaller segments of chromosomes, ranging in size from only a single gene up to a few percent of a chromosome's length, can cause complex birth defects like DiGeorge syndrome or even isolated autism without any obvious physical abnormalities. As a group, chromosome disorders are common, affecting approximately 7 per 1000 liveborn infants and accounting for approximately half of all spontaneous abortions occurring in the first trimester of pregnancy. These types of disorders are discussed in Chapter 6.

Single-gene defects are caused by pathogenic mutations in individual genes. The mutation may be present on both chromosomes of a pair (one of paternal origin and one of maternal origin) or on only one chromosome of a pair (matched with a normal copy of that gene on the other copy of that chromosome). Single-gene defects often cause diseases that follow one of the classic inheritance patterns in families (autosomal recessive, autosomal dominant, or X-linked). In a few cases, the mutation is in the mitochondrial rather than in the nuclear genome. In any case, the cause is a critical error in the genetic information carried by a single gene. Single-gene disorders such as cystic fibrosis (Case 12), sickle cell anemia (Case 42), and Marfan syndrome (Case 30) usually exhibit obvious and characteristic pedigree patterns. Most such defects are rare, with a frequency that may be as high as 1 in 500 to 1000 individuals but is usually much less. Although individually rare, single-gene disorders as a group are responsible for a significant proportion of disease and death. Overall, the incidence of serious single-gene disorders in the pediatric population has been estimated to be approximately 1 per 300 liveborn infants; over an entire lifetime, the prevalence of single-gene disorders is 1 in 50. These disorders are discussed in Chapter 7.

Multifactorial disease with complex inheritance describes the majority of diseases in which there is a genetic contribution, as evidenced by increased risk for disease (compared to the general public) in identical twins or close relatives of affected individuals, and yet the family history does not fit the inheritance patterns seen typically in single-gene defects. Multifactorial diseases include congenital malformations such as Hirschsprung disease (Case 22), cleft lip and palate, and congenital heart defects, as well as many common disorders of adult life, such as Alzheimer disease (Case 4), diabetes, and coronary artery disease. There appears to be no single error in the genetic information in many of these conditions. Rather, the disease is the result of the combined impact of variant forms of many different genes; each variant may cause, protect from, or predispose to a serious defect, often in concert with or triggered by environmental factors. Estimates of the impact of multifactorial disease range from 5% in the pediatric population to more than 60% in the entire population. These disorders are the subject of Chapter 8.

ONWARD

During the 50-year professional life of today's professional and graduate students, extensive changes are likely to take place in the discovery, development, and use of genetic and genomic knowledge and tools in medicine. Judging from the quickening pace of discovery within only the past decade, it is virtually certain that we are just at the beginning of a revolution in integrating knowledge of genetics and the genome into public health and the practice of medicine. An introduction to the language and concepts of human and medical genetics and an appreciation of the genetic and genomic perspective on health and disease will form a framework for lifelong learning that is part of every health professional's career.

GENERAL REFERENCES

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